

CSE 107 Fall 2026

Poisson calculations, Markov introduction, and final review

Week 10

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The increment model

Assume a homogeneous Poisson process with rate $\lambda > 0$ per hour.

- $N(0) = 0$; $N(t)$ counts arrivals in $(0, t]$.
- $N(b) - N(a) \sim \text{Pois}(\lambda(b - a))$ for $0 \leq a < b$.
- Counts on deterministic disjoint intervals are jointly independent.

Quantity	Meaning
λ	Rate: arrivals per hour
$\Delta = b - a$	Duration in hours
$m = \lambda\Delta$	Mean count, not a rate

Running example: $\lambda = 4$ per hour. These are modeling assumptions.

Reading: Harchol-Balter, pp. 216–217; DeGroot–Schervish, p. 293.

Translate the observation windows

At rate 4 per hour, define

$$A = N(1/4), \quad B = N(3/4) - N(1/4).$$



Count	Interval (hours)	Duration	Mean
A	$(0, 1/4]$	15 minutes	$4(1/4) = 1$
B	$(1/4, 3/4]$	30 minutes	$4(1/2) = 2$

A joint probability from independent counts

For disjoint intervals of lengths $\Delta_1, \Delta_2 > 0$, let A, B be their counts. For integers $a, b \geq 0$, start with

$$P(A = a, B = b).$$

Draw the intervals before using a product.

Two successive windows

At rate 4 per hour, find the probability of one arrival in the first 15 minutes and two in the following 30 minutes.

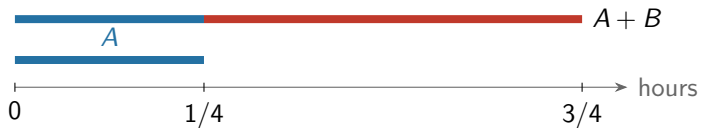


$$P(A = 1, B = 2) = ?$$

Cumulative counts share arrivals

The later cumulative count contains the earlier one:

$$N(3/4) = \underbrace{N(1/4)}_A + \underbrace{N(3/4) - N(1/4)}_B.$$



The joint law of two cumulative counts

Let $0 < s < t$. Set $A = N(s)$ and $B = N(t) - N(s)$, so $A \perp\!\!\!\perp B$.

$$\{N(s) = i, N(t) = j\} = \{A = i, B = j - i\}.$$

Support check: $i \geq 0$ and $j - i \geq 0$. A cumulative count cannot decrease.

A future count after an observed count

For $0 < s < t$, condition on $N(s) = i$. The new count is $B = N(t) - N(s)$.

$$P(N(t) = j \mid N(s) = i) = P(B = j - i \mid A = i).$$

The new increment is independent of the past count; the total $N(t)$ contains it.

Covariance of cumulative counts

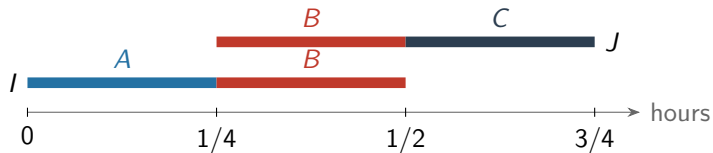
Write $N(s) = A$, $N(t) = A + B$, where $A \perp\!\!\!\perp B$.

$$\text{Cov}(N(s), N(t)) = \mathbb{E}[A(A + B)] - \mathbb{E}[A]\mathbb{E}[A + B].$$

The shared arrivals contribute to both counts. Correlation requires positive variances.

Partially overlapping windows

At rate 4 per hour, count arrivals in $I = (0, 1/2]$ and $J = (1/4, 3/4]$.



$$K_I = A + B, \quad K_J = B + C; \quad A, B, C \text{ independent Pois}(1).$$

Counts and covariance

Arrivals form a homogeneous Poisson process at rate 6 per hour. Let U count the first ten minutes and V the first thirty minutes.

- 1 Write each count using disjoint pieces.
- 2 Find $\text{Cov}(U, V)$. Are U and V independent?

Check the interval before multiplying

Assume the same homogeneous Poisson process, and $0 < s < t$.

Claim

Verdict and reason

$N(s) \perp\!\!\!\perp [N(t) - N(s)]$.

$N(s) \perp\!\!\!\perp N(t)$.

Two equal-length intervals have identical counts.

Specify the windows, isolate disjoint pieces, then choose the probability rule.

A state observed each day

Let X_n be a machine's state at daily check n .

State	Meaning	Observation step
0	Working	One daily check to the next
1	Under repair	The same observation schedule

Model the next state using the current state. Specify the starting state or its probability distribution separately.

All Markov-chain material is excluded from the final examination.

The Markov assumption

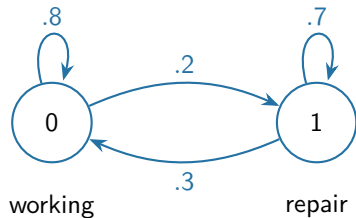
For a history with positive probability, the **Markov property** says

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i).$$

Time homogeneity adds that the last probability is a fixed p_{ij} , independent of n .

Harchol-Balter p. 421; DeGroot–Schervish pp. 188–191.

From a diagram to a matrix



$$P = \begin{array}{c|cc} & 0 & 1 \\ \hline 0 & .8 & .2 \\ 1 & .3 & .7 \end{array}$$

Row: present state.

Column: next state.

$$p_{ij} = P(X_{n+1} = j \mid X_n = i).$$

State order is $(0, 1)$ throughout. One step = one daily check.

Each row is a conditional PMF

Given state i , the next state lies in the finite state set S .

$$p_{ij} \geq 0, \quad \sum_{j \in S} p_{ij} = 1.$$

A row is one conditional distribution. A column is not.

One-step distribution update

Define $\pi_{n,i} = P(X_n = i)$. To find the next-state probability, partition by the present state.

$$\pi_{n+1,j} = P(X_{n+1} = j).$$

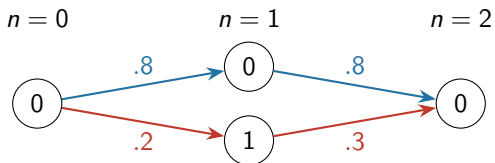
A random initial state

Suppose $\pi_0 = (1/4, 3/4)$, in state order $(0, 1)$.

$$P = \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix}, \quad \pi_1 = \pi_0 P.$$

Two routes to a two-step return

Start at $X_0 = 0$. Find $P(X_2 = 0 \mid X_0 = 0)$ by conditioning on X_1 .



DeGroot–Schervish p. 194: conditioning on the intermediate state.

Why the matrix is squared

For any start i and destination j , sum over the intermediate state k :

$$P(X_2 = j \mid X_0 = i) = \sum_k p_{ik} p_{kj} = (P^2)_{ij}.$$

The updated law is a probability law

Assume $\pi_{n,i} \geq 0$, $\sum_i \pi_{n,i} = 1$, and each row of P sums to 1.

$$\pi_{n+1,j} = \sum_i \pi_{n,i} p_{ij}.$$

One and two steps from repair

The machine is under repair now: $X_0 = 1$.

$$P = \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix}.$$

Find the probability it is working (a) at the next check and (b) two checks later.

What the model assumes

Check	Meaning
State description	Does the current state contain the information needed for the next transition?
Markov property	Given that state, earlier history does not change the next-state law.
Time homogeneity	The same transition probabilities apply at each daily check.

Required lesson ends at one- and two-step probabilities. All chains are excluded from the final.

Final scope

Required core	Events and conditioning; random variables and transformations; joint laws and moments; bounds, LLN and CLT; known-SD mean intervals; Poisson counts and increments.
Excluded	All Markov-chain material; optional extensions marked in the weekly materials.
Preparation	Use the ungraded review packet and its independent worked answers. Choose additional exercises by topic.

Markov's inequality remains in the bounds material. It is not a Markov-chain result.

Use the official course announcement for the final's confirmed date, time, room, and working duration.

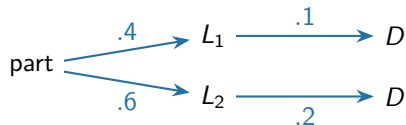
Choose the probability statement

Type	Symbol	What the model supports
Exact calculation	$=$	A probability from the specified law
Bound	$\leq$	A guaranteed upper bound, often using moments
Limit	$\longrightarrow$	Behavior as $n \rightarrow \infty$, with tolerance fixed
Approximation	$\approx$	A substitute law, such as a normal approximation
Interval coverage	$1 - \alpha$	Probability that a random interval covers a fixed parameter

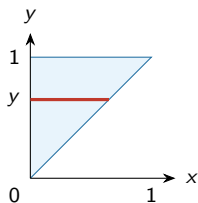
Write the claim your assumptions justify. A bound is not an exact tail probability.

Conditioning reverses the question

Two production lines supply 40% and 60% of the parts. Their defect rates are 10% and 20%. Let D denote a defect.



Support determines the conditional law



The familiar joint density is $f_{X,Y}(x,y) = 2$ on $0 < x < y < 1$, and zero otherwise.

At fixed $0 < y < 1$, which x values are allowed?

Transform an event, then differentiate

In the same triangle model, $f_Y(y) = 2y$ on $(0, 1)$. Set $Z = Y^2$.

$$F_Z(z) = P(Y^2 \leq z).$$

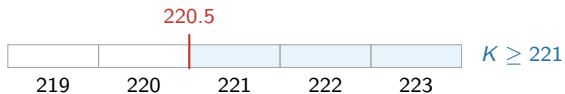
First determine the transformed support and all CDF branches.

A guarantee and a limiting statement

Let X_i be iid with mean μ and variance 4. How large is sufficient for $P(|\bar{X}_n - \mu| \geq .5) \leq .1$?

A discrete tail and its normal approximation

Let $K \sim \text{Bin}(400, .5)$. Find an exact expression and a normal approximation for $P(K > 220)$.



An interval for a mean

For iid normal observations with unknown mean μ and known SD 4, suppose $n = 64$ and the observed mean is 12. Start from $Z = (\bar{X}_n - \mu)/(4/\sqrt{64}) \sim N(0, 1)$.

Convert cumulative counts to increments

Arrivals form a homogeneous Poisson process at rate 3 per hour. Find

$$P(N(1/3) = 0, N(1) = 2).$$

Check a complete solution

Repair each claim and state the missing reason.

- ① For iid data of variance σ^2 , $\text{Var}(\bar{X}_n) = \sigma^2/\sqrt{n}$.
- ② In a rate- $\lambda > 0$ Poisson process, $N(s) \perp\!\!\!\perp N(t)$ whenever $0 < s < t$.

Use the review answers to check assumptions, support, event translation, scale, and conclusion.

Optional: joint counts with overlap

Let independent $A \sim \text{Pois}(a)$, $B \sim \text{Pois}(b)$, $C \sim \text{Pois}(c)$, and set $X = A + B$, $Y = B + C$. Write $p_m(k)$ for a Poisson PMF. For integers $x, y \geq 0$:

Unassessed extension. The required lesson uses only the shared-count covariance.

Optional: an earlier count given a total

For $0 < s < t$, condition on $N(t) = n$. Use a joint-to-marginal PMF ratio.

Unassessed extension. Conditioning on the total is different from conditioning on the past.

Optional: the first-arrival connection

Let $T_1 = \inf\{t : N(t) \geq 1\}$ be the time of the first arrival.

$$\{T_1 > t\} = \{N(t) = 0\}, \quad t \geq 0.$$

Unassessed extension. No arrival-time question appears in the required final.

Optional: a larger two-step example

Use states (1, 2, 3, 4) and the transition matrix

$$P = \begin{pmatrix} 1/4 & 1/4 & 0 & 1/2 \\ 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}.$$

Find the two-step probability from state 3 to state 2.

Adapted from supplied STAT 131 problems, p. 13, §5, Problem 1(a). Optional; excluded from the final.

Selected reading

Poisson increments

Harchol-Balter, supplied 2024 text, pp. 216–217. Alternative: DeGroot–Schervish, 4th ed., p. 293, Definition 5.4.2 and Example 5.4.6 only.

Markov introduction — excluded from the final

Harchol-Balter p. 421 through the row-sum discussion on p. 422; only the first repair-model matrix/diagram on p. 423. DeGroot–Schervish pp. 188–191 definitions and p. 194 two-step derivation.

Final review

Reuse the selected covariance, bounds, CLT, and mean-interval readings from Weeks 6–9. No new whole-chapter assignment.

Optional

Harchol-Balter p. 218 (first-arrival connection); supplied STAT 131 problems p. 13, §5, 1(a).
Printed page numbers. “Time homogeneity” does not assert a stationary marginal distribution.